

Customer Churn Prediction & Business Intelligence System

Professional Business & Data Science Report

This end-to-end system predicts customer churn, identifies churn drivers, estimates financial impact, and prioritizes retention actions. The current reproducible experiment uses 5,000 synthetic customer records covering telecom/SaaS-style customer, account, service, financial, usage, support and payment attributes.

Business Questions

- Which customers are likely to leave?
- What factors contribute to churn?
- Which segments have the highest churn?
- What is the estimated financial impact?
- Which customers should be prioritized for retention?

Data Science Workflow

Raw Data → Cleaning → EDA → Feature Engineering → Train/Test Split → Model Training → Evaluation → Explainability → Prediction → Dashboard

Model Comparison — Actual Experiment Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression	0.712	0.332	0.678	0.446	0.769	0.448
Gradient Boosting	0.839	0.589	0.193	0.291	0.743	0.422
Random Forest	0.815	0.450	0.368	0.405	0.742	0.382
XGBoost	0.833	0.532	0.193	0.283	0.722	0.399
Decision Tree	0.727	0.328	0.567	0.415	0.680	0.325

Best model by ROC-AUC: Logistic Regression (ROC-AUC 0.769).

Business Feature Engineering

- Customer_Lifetime_Months
- Average_Monthly_Spend
- Support_Calls_Per_Month
- Payment_Delay_Rate
- Service_Count
- Usage_Change_Percentage

Explainability & Retention

The dashboard combines permutation importance with customer-level business rules. High-risk customers are prioritized using churn probability and estimated annual revenue at risk. Recommended actions include contract interventions, proactive support, billing/payment interventions, and targeted engagement.

Technology Stack

Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Scikit-learn, XGBoost, SHAP-ready explainability, SQL/PostgreSQL, Streamlit, Power BI, GitHub and cloud deployment.

Portfolio Outcome

This project demonstrates the complete path from data cleaning and EDA through feature engineering, machine learning, evaluation, explainable AI, business analytics, dashboarding and deployment.